

Marchal's family: Inclined co-orbitals in the three-body problem

I: Stability of inclined co-orbital planetary systems

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Introduction

The Lagrange relative equilibrium

Lagrange relative equilibrium of the
3BP

Elliptic if

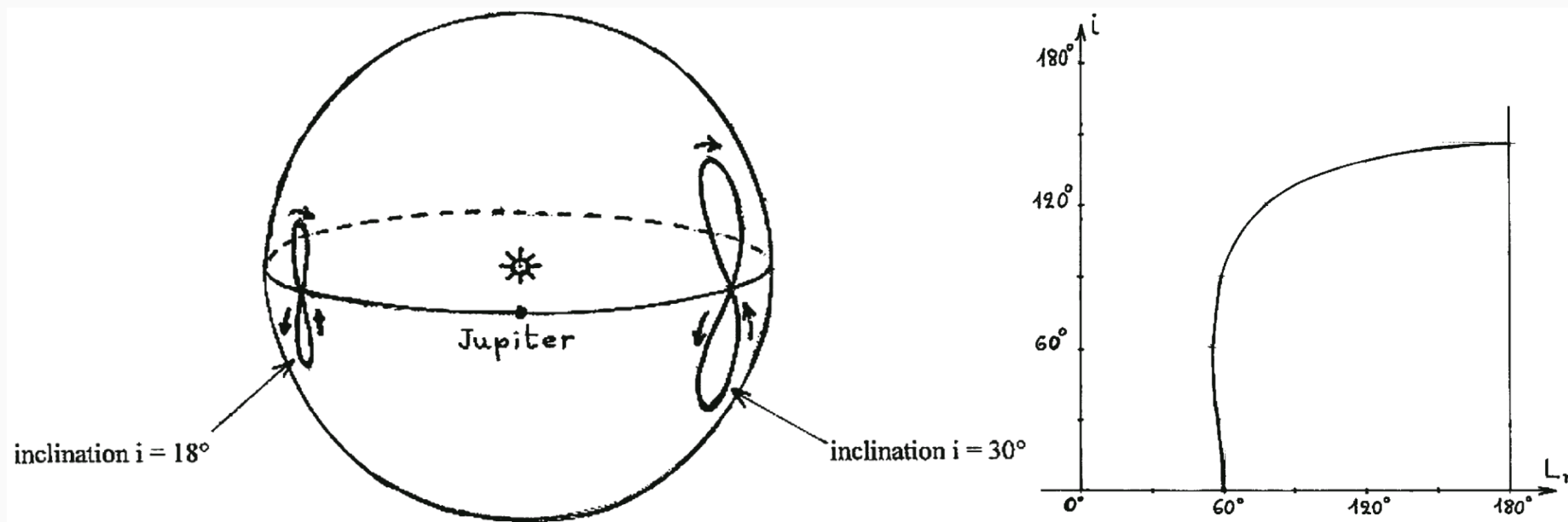
$$\frac{m_0 * m_1 + m_0 * m_2 + m_1 * m_2}{(m_0 + m_1 + m_2)^2} > \frac{1}{27}$$

$$\begin{cases} (1 - \varepsilon, \varepsilon, 0) \rightarrow \varepsilon_G \approx 0.0385 \\ (1 - 2\varepsilon, \varepsilon, \varepsilon) \rightarrow \varepsilon_G \approx 0.0191 \end{cases}$$

The Lagrange relative equilibrium – Lyapunov families

Lyapunov families emerging from the elliptic directions of Lagrange 3 planar Lyapunov families... One non-planar : the **vertical family**

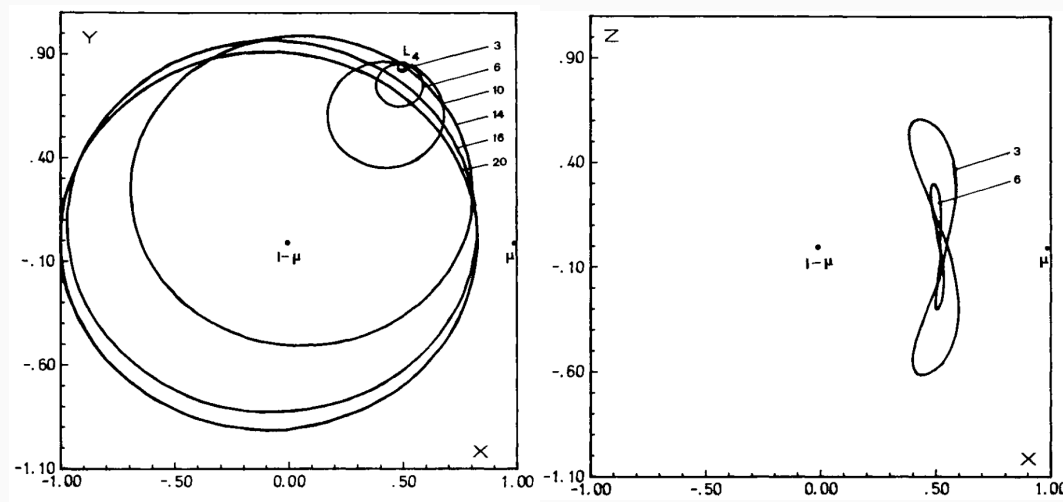
The vertical family – small masses



- (Marchal, 2009) – $\mathcal{VF}_L$ in the average restricted problem
 - Already a conjecture – part of a three-parameter family, including P_{12} !
- (Leleu, 2016) – numerical hints of a stable region in the full planetary problem

A historical note: forgetting previous work

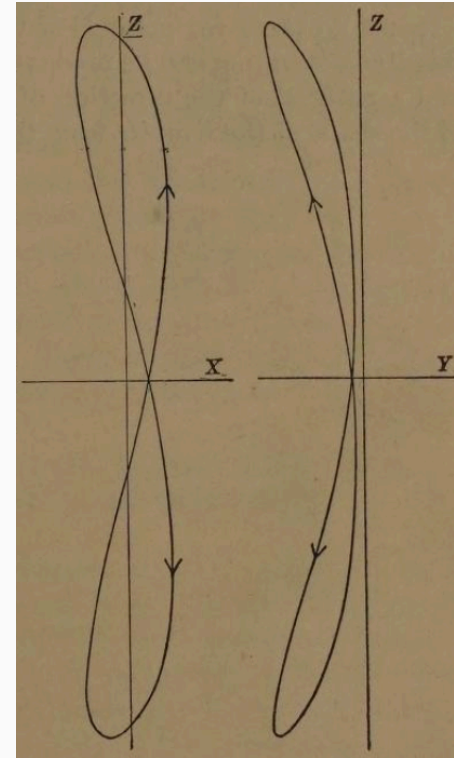
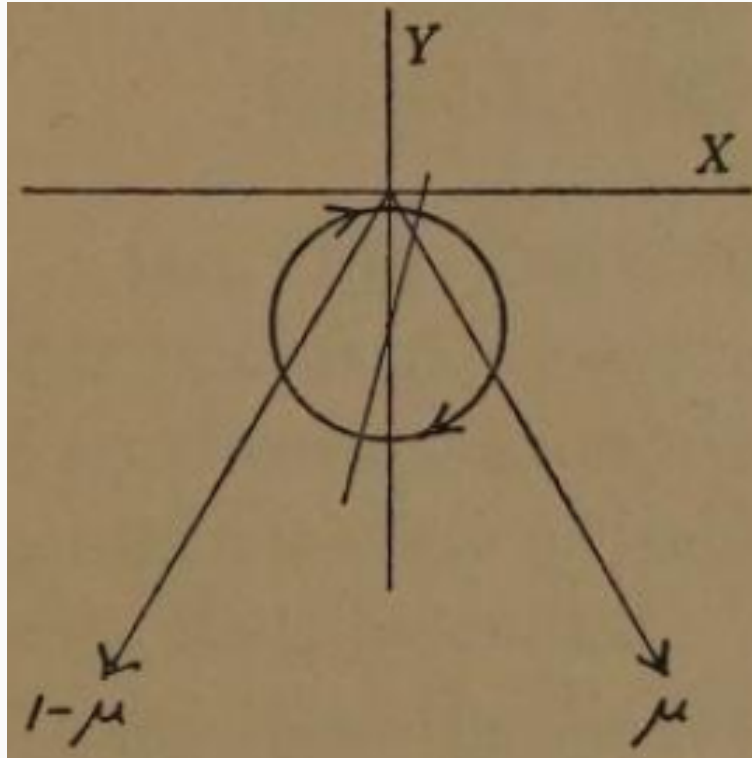
(Marchal, 2009) results are in (Zagouras, 1985)



But (Zagouras, 1985)'s method is described in “Zagouras, C.G.: 1982, Post doctoral dissertation, University of Patras, Greece.”

A historical note: forgetting previous work

... And (Zagouras, 1985) is an extension of (Moulton, 1920), chap IX (by Buck)!



Questions

- Q1: Does the family persist in the non-restricted/non-average problem?
- Q2: What is the stability of the orbits along this family?
- Q3: Does it have an impact on the global stability of the co-orbital region?

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$\mathcal{VF}_L$ in the average circular
planetary problem

Frame of reference

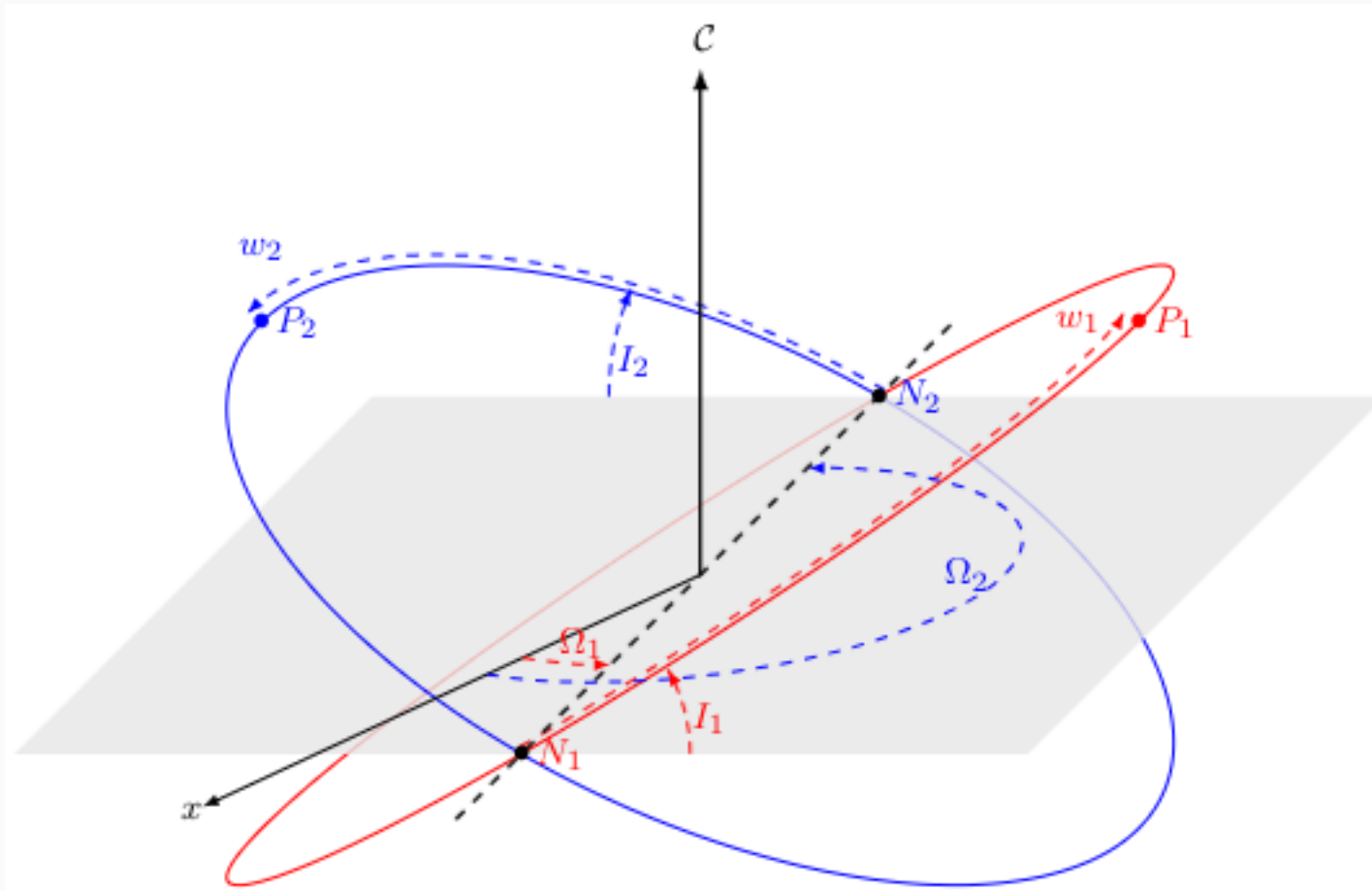
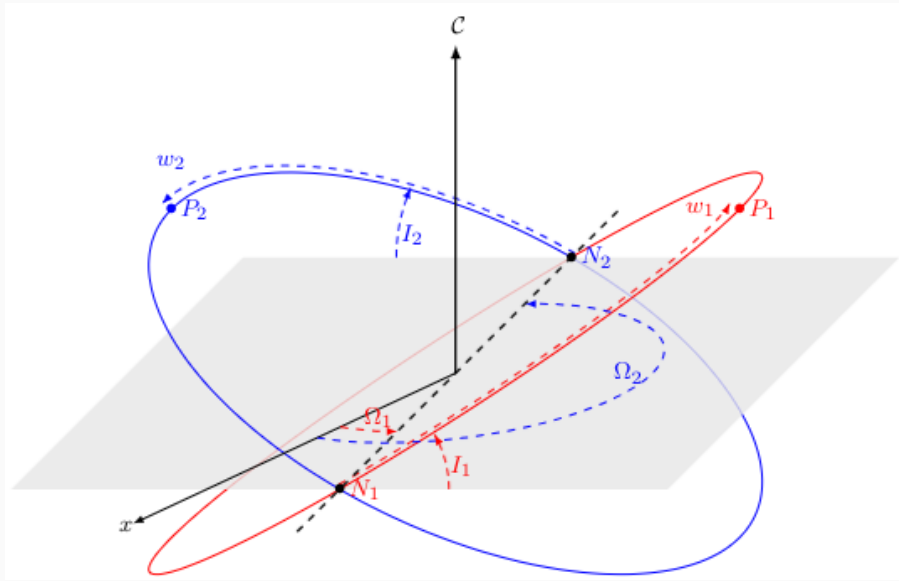


Figure 4: Canonical heliocentric : heliocentric positions, barycentric momenta; $(Oxy) \perp C$
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Complex Poincaré-like coordinates



$$\beta_j = \frac{m_0 m_j}{m_0 + m_j}, \mu_j = \mathcal{G}(m_0 + m_j)$$

$$\Lambda_j = \beta_j \sqrt{\mu_j} \mathbf{a}_j$$

$$\tilde{\lambda}_j = \mathbf{M}_j + \omega_j$$

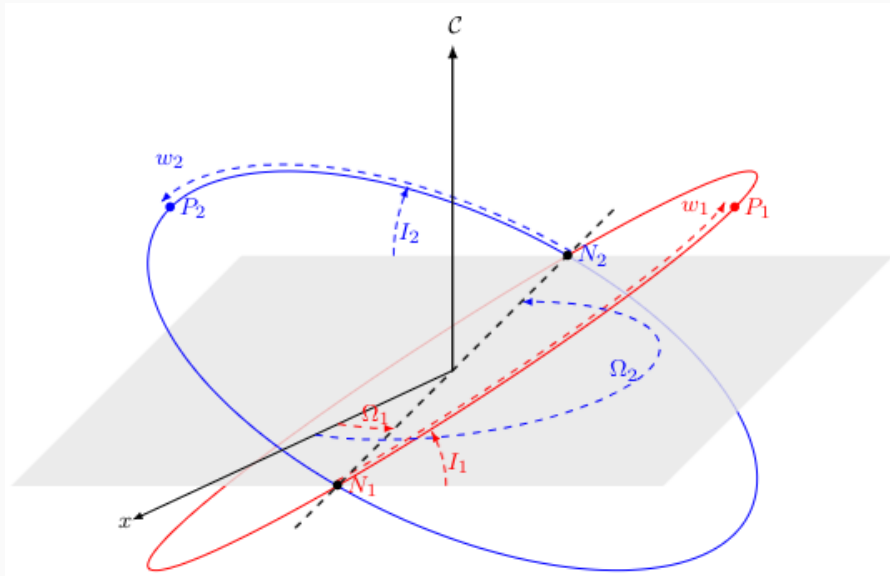
$$\mathbf{x}_j = \underbrace{\sqrt{\Lambda_j - \mathbf{G}_j}}_{\sim \mathbf{e}_j} \exp(i \omega_j)$$

$$\tilde{\mathbf{x}}_j = -i \bar{\mathbf{x}}_j$$

$$\Phi_j = \mathbf{G}_{1_z} \pm \mathbf{G}_{2_z}$$

$$\varphi_j = \frac{\Omega_1 \pm \Omega_2}{2}$$

Jacobi reduction



$$\Omega_2 - \Omega_1 = \pi = 2\varphi_2$$

$\Rightarrow \varphi_1$ and Φ_1 ignorable

$$\Phi_1 = C_z = C = \text{constant}$$

$\Rightarrow \varphi_2$ ignorable

Jacobi reduction – consequences

We are left with 4 degrees of freedom and one parameter C

As we do not track the nodes, they are at a fixed position in this frame:
draconic frame

→ We haven't lost the nodes or the inclinations!

$$\left\{ \begin{array}{l} J := I_1 + I_2, \quad \cos(J) = \frac{C^2 - G_1^2 - G_2^2}{2G_1G_2}, \quad C \cos(I_j) = G_j + G_{1-j} \cos(J) \\ \dot{\Omega}_j = \frac{\partial H}{\partial C} \end{array} \right.$$

Co-orbital motion in the planetary case

Planetary problem

$$(1 - \varepsilon(m_1 + m_2), \varepsilon m_1, \varepsilon m_2), \varepsilon \ll 1$$

ε -neighborhood of **co-orbital** motions

$$a_1, a_2 = a^* + \mathcal{O}(\varepsilon)$$

Averaging over fast angle ζ_2

The submanifold of eccentricity 0 is invariant under the flow: **circular problem**

A one-parameter family: $\mathcal{V}\mathcal{F}_L$

$$\bar{F}(Z_1, \zeta_1, C) = \alpha Z_1^2 + \varepsilon \beta F_1(\zeta_1, C)$$

For a fixed C_0 , one-parameter family of equilibrium points satisfying

$$Z_1 = 0, \quad \zeta_1 = \zeta(C_0), \quad \frac{\partial F_1}{\partial \zeta_1}(\zeta(C_0), C_0) = 0$$

F_1 independant of masses! $\Rightarrow$ Study valid for all small masses

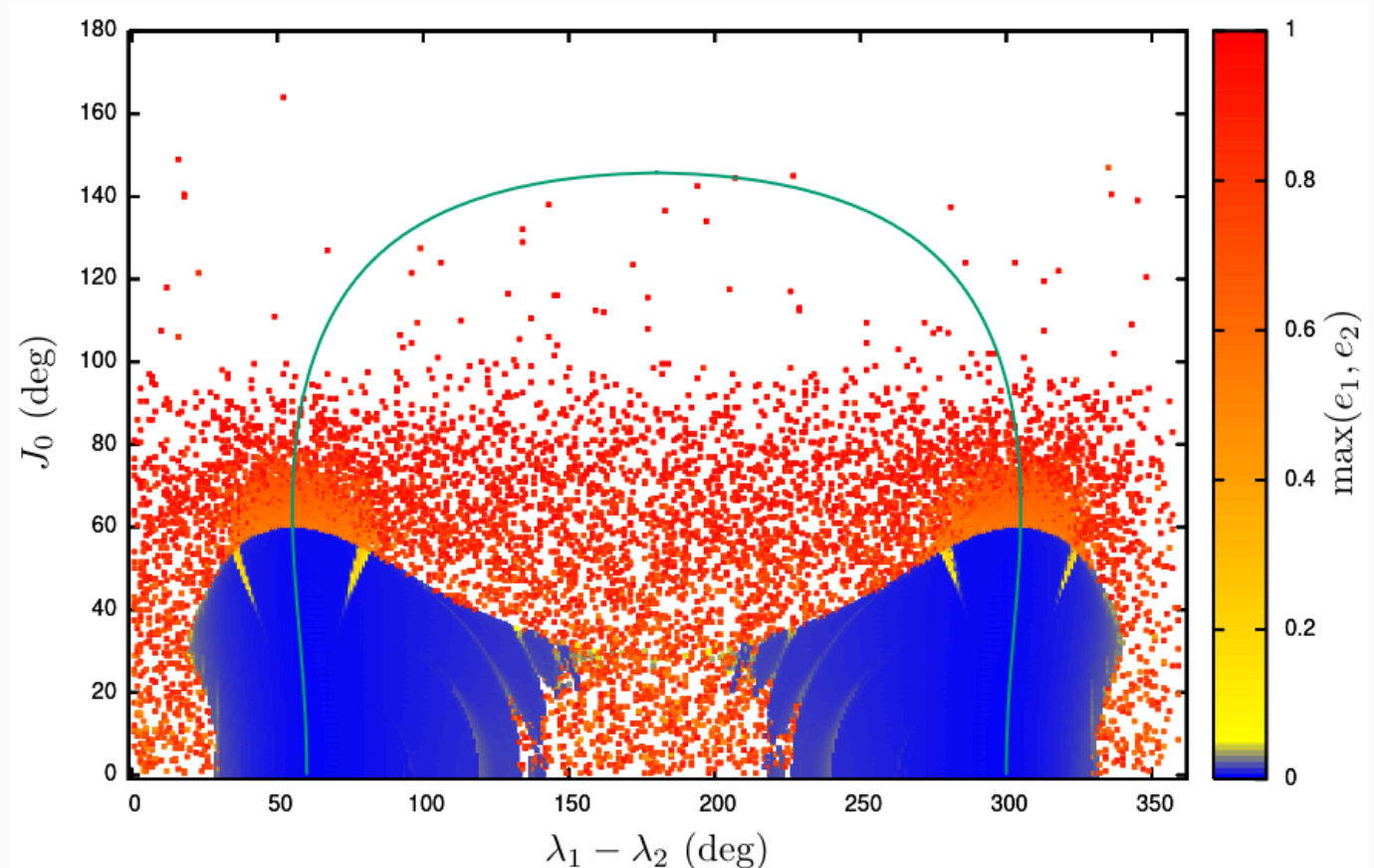
Next ?

- Analytical approximation
- Or numerical search

What can we say?

Analytical: 20th degree expansion in $\sin\left(\frac{J_{\max}}{2}\right)$

Stability: $\mathcal{V}\mathcal{F}_L$ is stable
in the average **circular**
problem



Numerical search in the full problem

A new set of coordinates

- Numerical integration: need coordinates with **explicit** expression of the Hamiltonian/vector field
- The Jacobi reduction should be explicit in these coordinates

Hill coordinates!

$$\begin{cases} R_j \\ G_j \end{cases} \quad \begin{matrix} r_j \\ w_j \end{matrix}$$

A new set of coordinates

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Hill coordinates!

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Hill coordinates!

$$\begin{cases} R_j & r_j \\ G_j & w_j \\ \cancel{G_j \cos(l_j)} & \cancel{\Omega_j} \end{cases}$$

How to search for periodic orbits?

Looking for zeros of

$$f : \begin{cases} \mathbb{R}^n & \rightarrow \mathbb{R}^n \\ (\mathbf{x}; C) & \mapsto \Phi_T(\mathbf{x}; C) - \mathbf{x} \end{cases}$$

But the zeros aren't locally unique!

Solution: add **sections**

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How to search for periodic orbits?

Looking for zeros of

$$f : \begin{cases} \mathbb{R}^n & \rightarrow \mathbb{R}^n + m \\ (\mathbf{x}; C) & \mapsto \begin{pmatrix} \Phi_T(\mathbf{x}; C) - \mathbf{x} \\ \sigma(\mathbf{x}; C) \end{pmatrix} \end{cases}$$

But the zeros aren't locally unique!

Solution: add **sections**

How to search for periodic orbits?

In our case:

$$\begin{cases} \sigma_1(\mathbf{x}; C) = w_1 + w_2 \\ \sigma_2(\mathbf{x}; C) = J - J_0 \end{cases} \quad \text{Selecting a point on the orbit}$$

Focusing on the case of **two equal masses**: $(1 - 2\varepsilon, \varepsilon, \varepsilon)$

Rootfinding algorithm: multi-dimensional **Newton-Raphson** method

Numerical integration with `taylor` software (**Jorba and Zou, 2005**) via

`TaylorInterface.jl` (in **julia** (**Bezanson et al., 2017**))

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Selecting an orbit in the family

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Small masses: $\varepsilon = 10^{-3}$

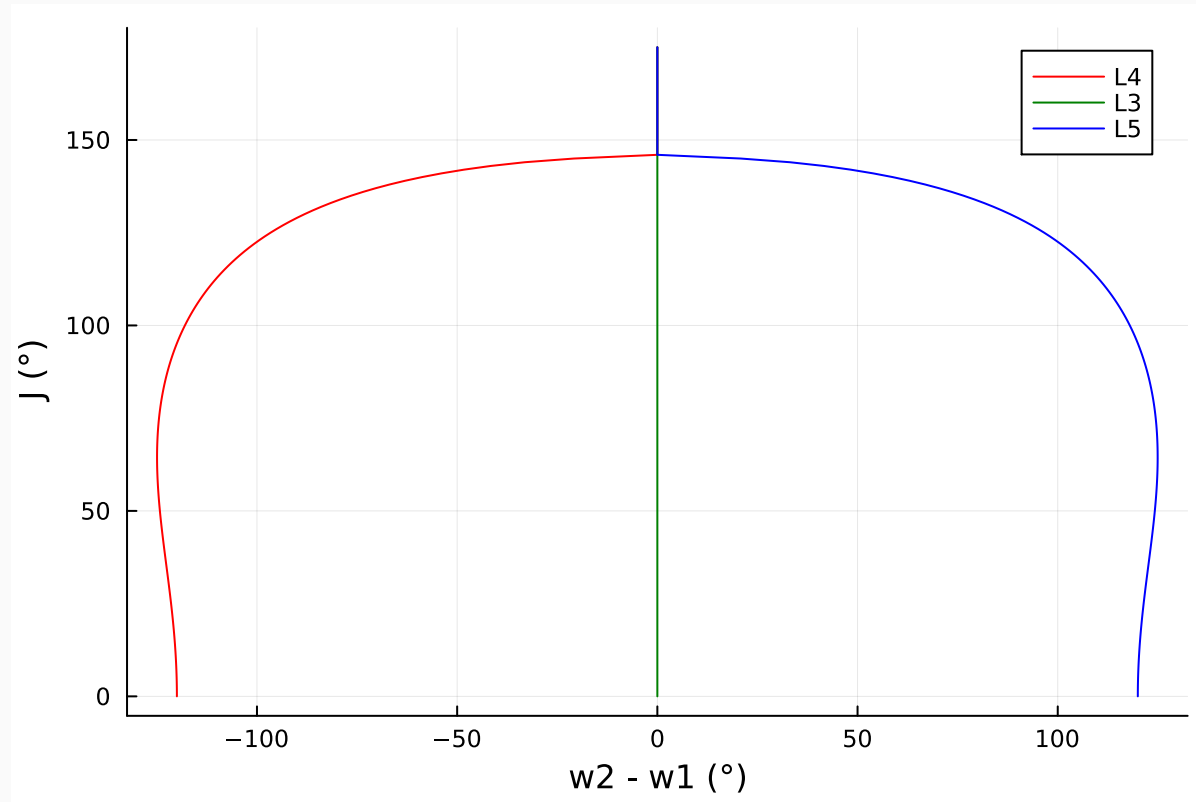
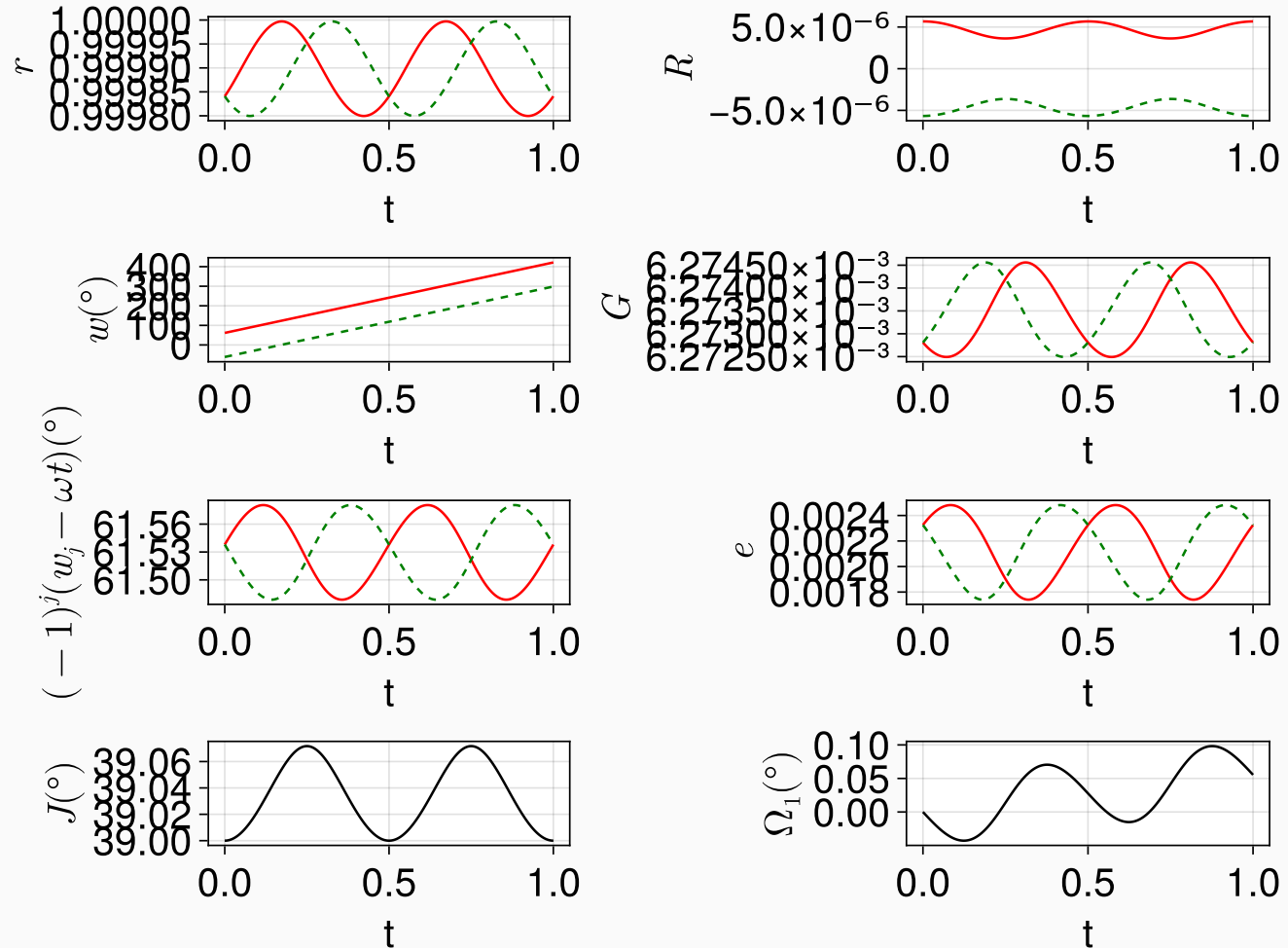


Figure 8: $\mathcal{VF}_L$ from L_4 and L_5 , $\mathcal{VF}_E$ from L_3

Coordinates along trajectory at $\varepsilon = 10^{-3}$

Coordinates at $J = 39^\circ$



Trajectory examples

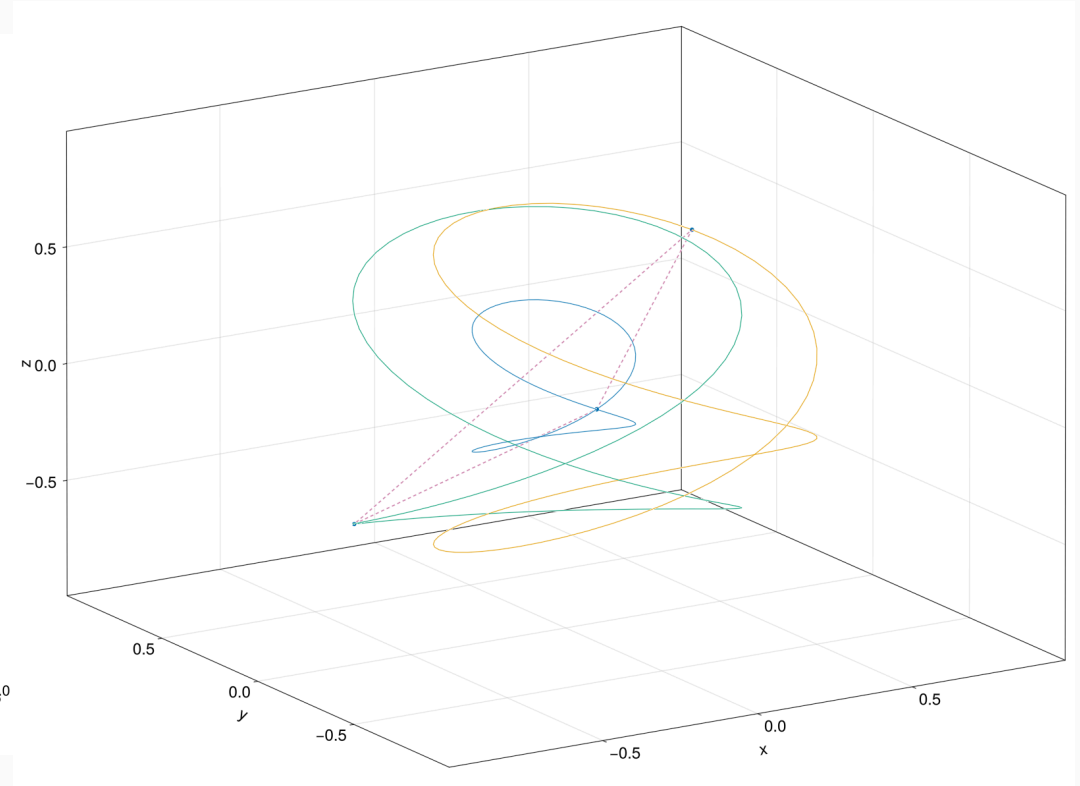
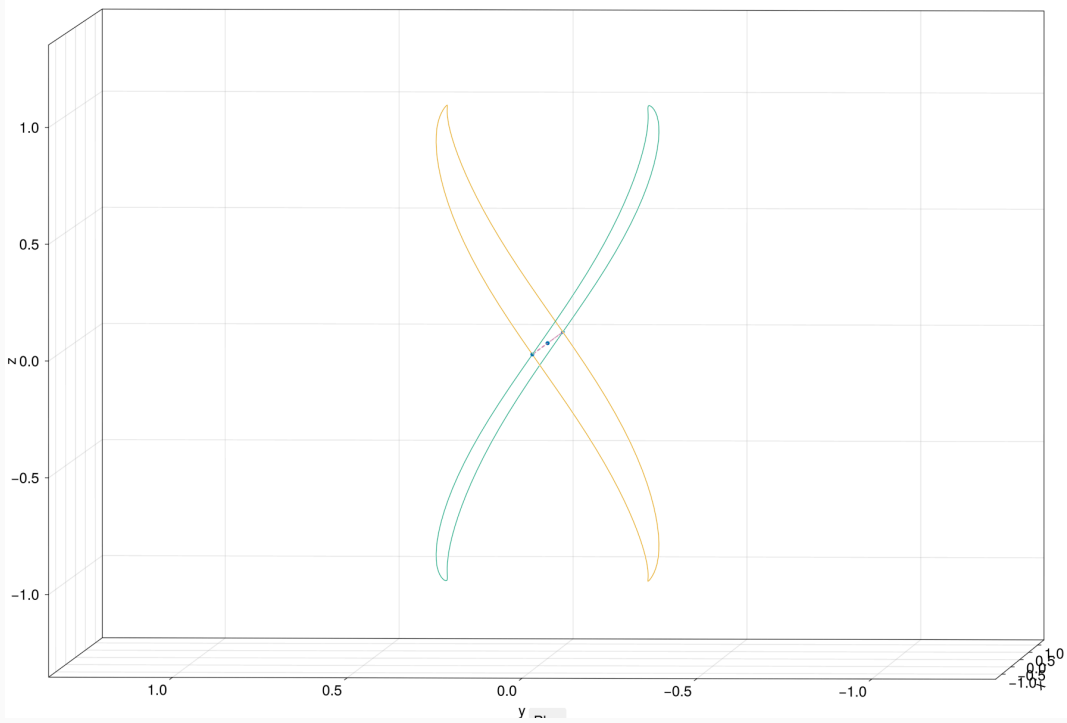
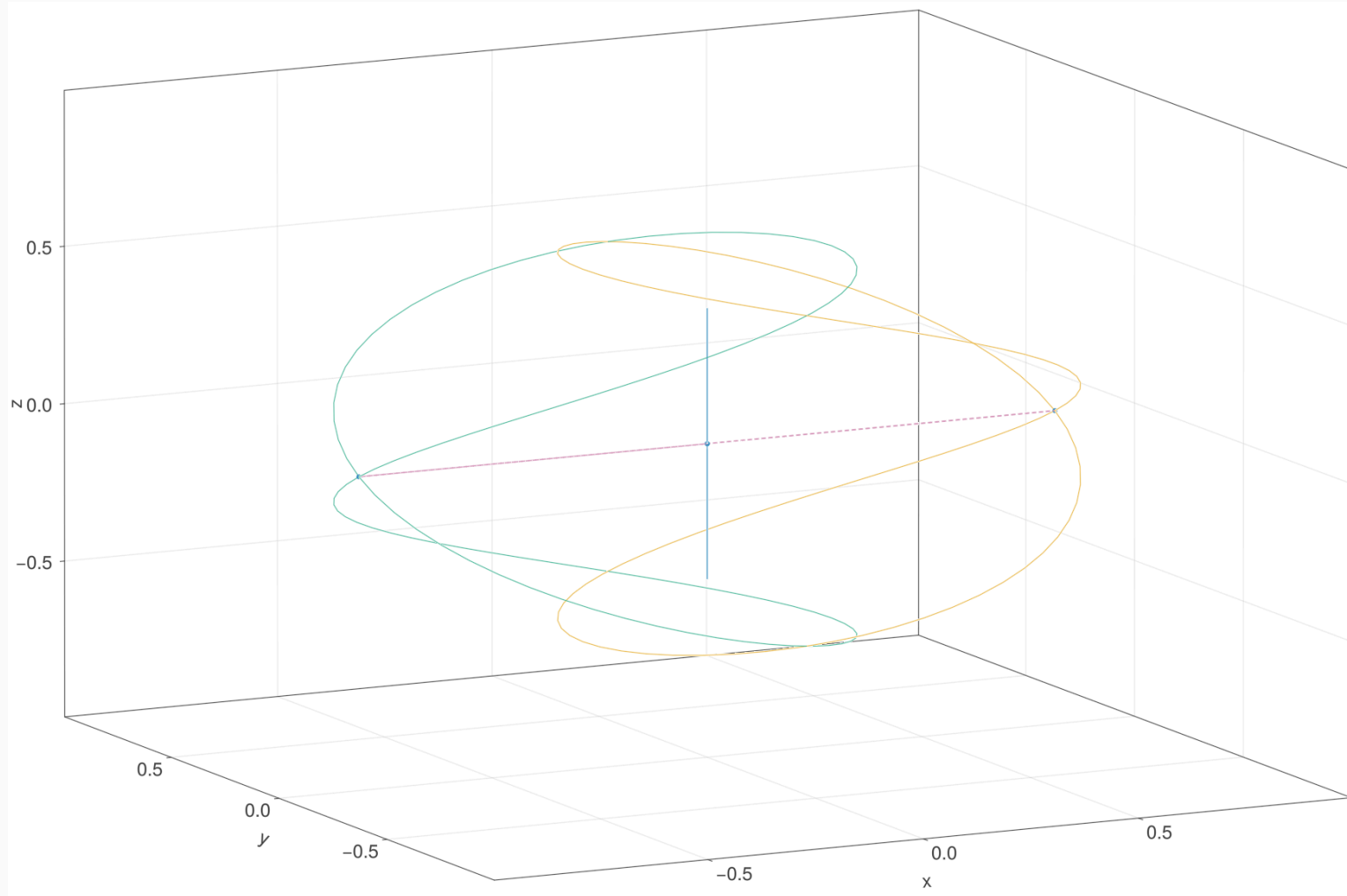


Figure 10: $\mathcal{VF}_L$ orbits in the draconic (left) and uniformly rotating (right) plane, for $\varepsilon = 0.21, J = 106^\circ$.

Trajectory example – $\mathcal{V}\mathcal{F}_E$

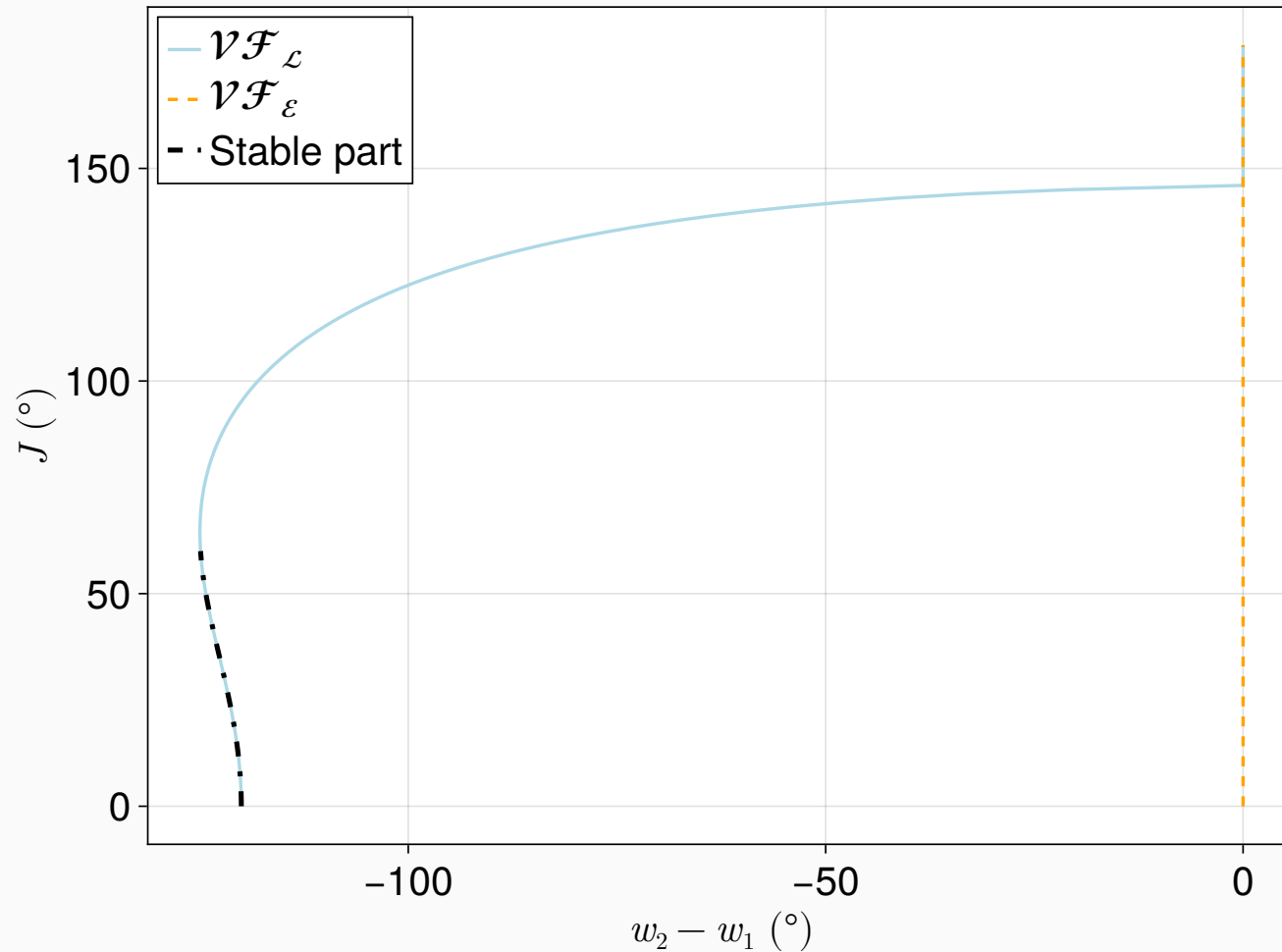


CM&DA webinar 2026 – Alexandre Figure 11: $\mathcal{V}\mathcal{F}_E$ orbit at $\varepsilon = 0.21, J = 130^\circ$.

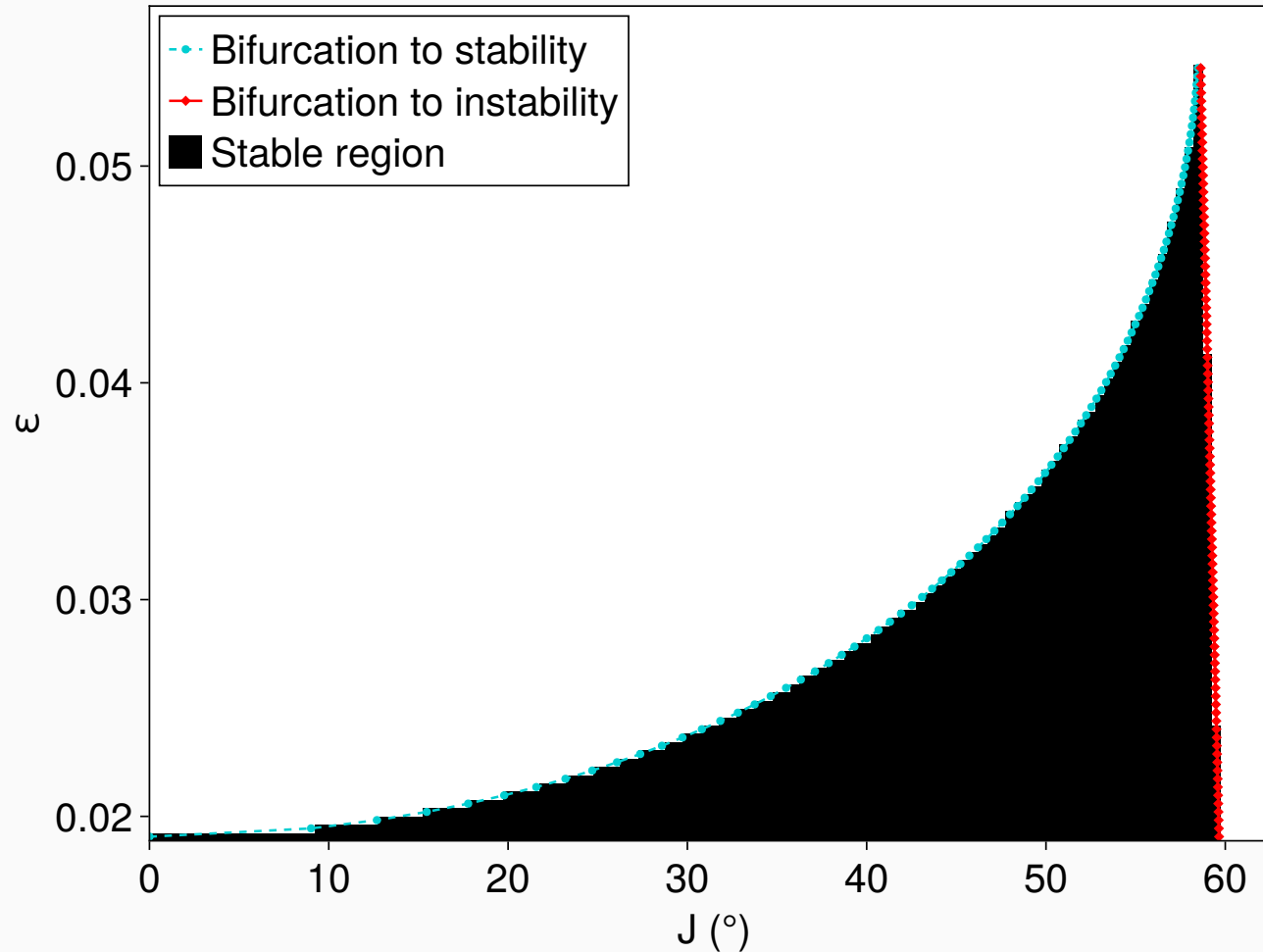
Stability



Stability at $\varepsilon = 10^{-3}$

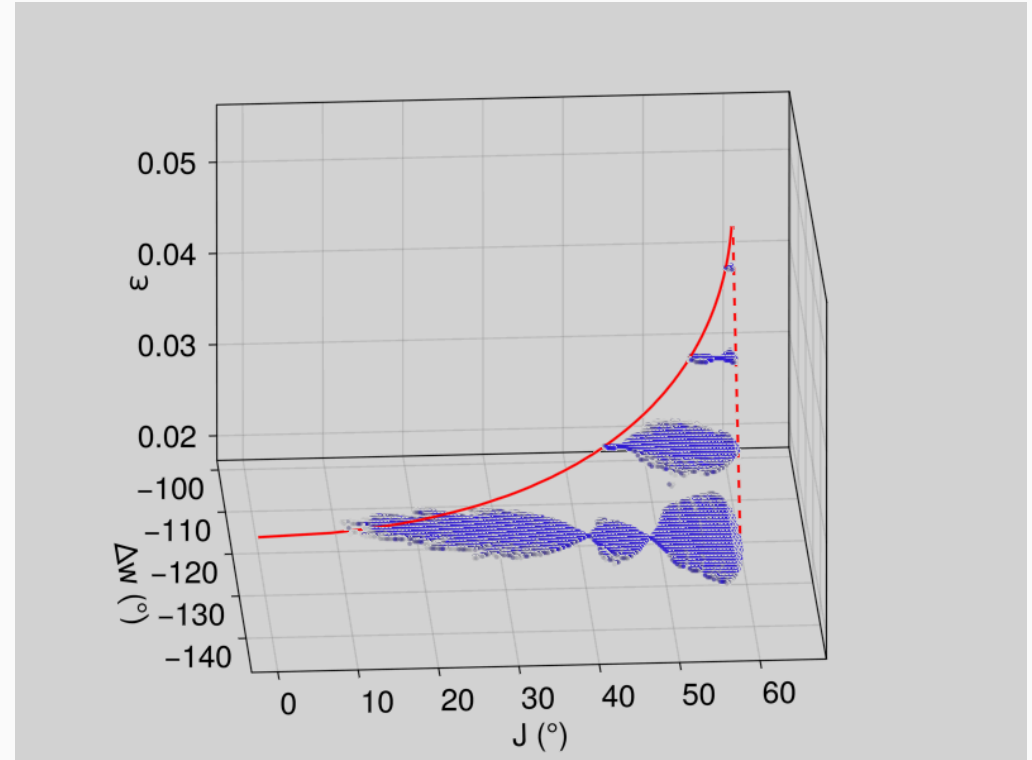
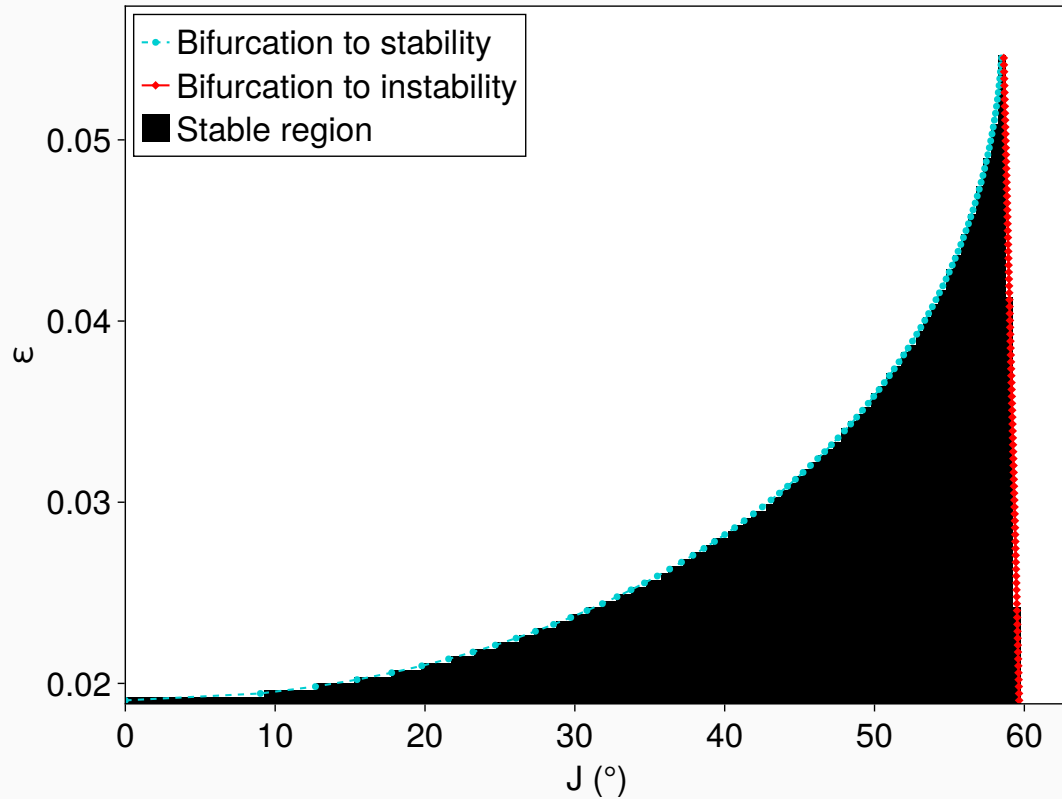


Increasing masses: above Gascheau's value

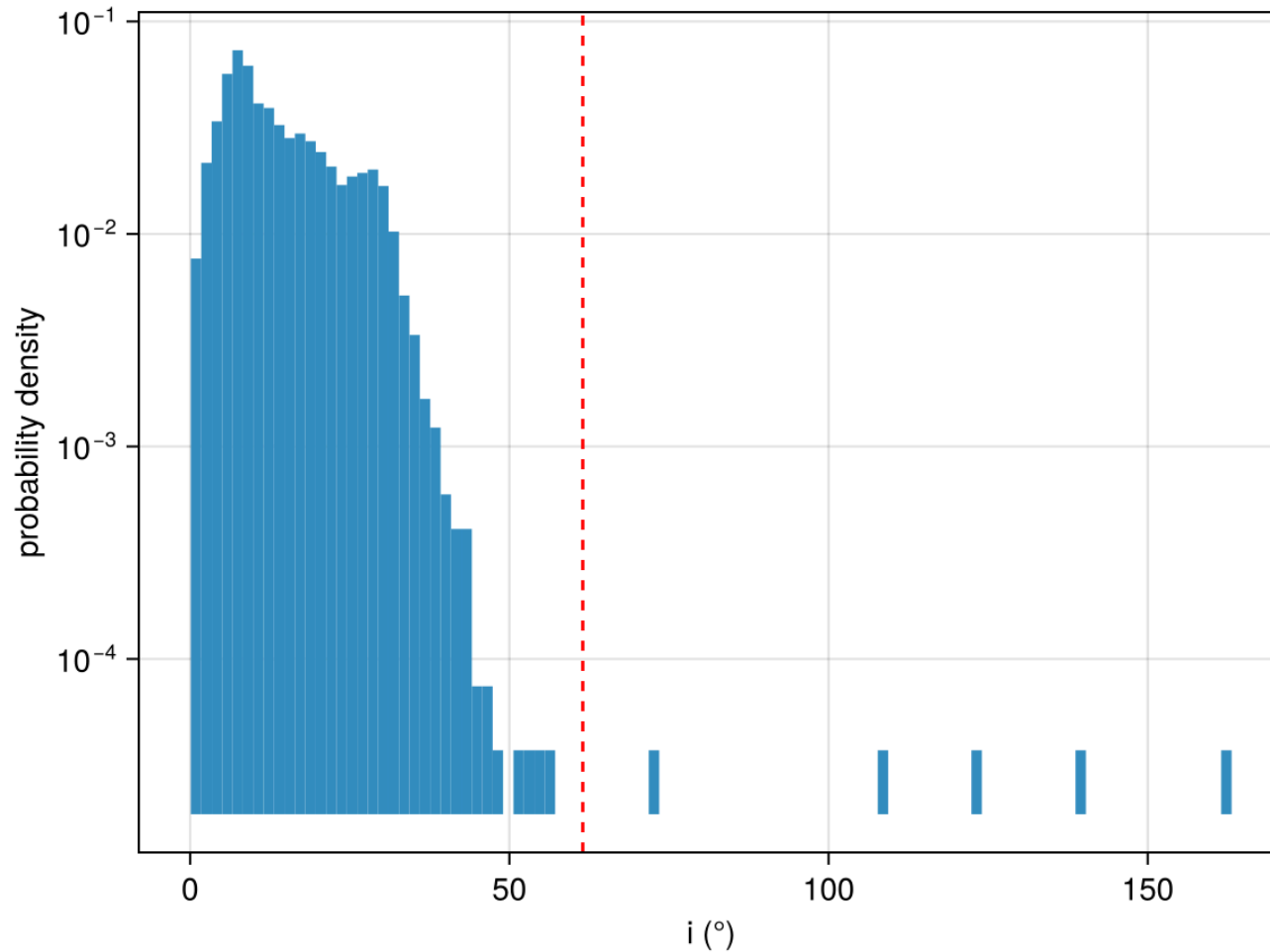


Similar to (Roberts, 2002)

Above Gascheau's value (long-term stability)



Astronomical consequences?



Conclusion

Answering our initial questions

Q1: Does the family persist in the non-restricted/non-average problem?

- Yes

Q2: What is the stability of the orbits along this family?

- Stability changes, stability remnants beyond Gascheau

Q3: Does it have an impact on the global stability of the co-orbital region?

- Yes

Future work

- Link $\mathcal{VF}_L$ to the vertical family at equal masses, P_{12}
 - Paradox: P_{12} leads to the Eight, $\mathcal{VF}_L$ ends at collisions
 - Study of $\mathcal{VF}_L$ symmetries
- (Zagouras, 1985) mentioned a bifurcation, and this would explain the endpoint divergences between $\mathcal{VF}_L$ and P_{12}
- Apply `TaylorInterface.jl` and our numerical toolbox to more problems

Thank you for your attention!

Bibliography

- Bezanson, J. et al. (2017) “Julia: A Fresh Approach to Numerical Computing,” *SIAM review*, 59(1), pp. 65–98.
- Jorba, À. and Zou, M. (2005) “A Software Package for the Numerical Integration of ODEs by Means of High-Order Taylor Methods,” *Experimental Mathematics*, 14(1), pp. 99–117. Available at: <https://doi.org/10.1080/10586458.2005.10128904>.
- Leleu, A. (2016) *Dynamique Des Planètes Coorbitales*.
- Marchal, C. (2009) “Long Term Evolution of Quasi-Circular Trojan Orbits,” *Celestial Mechanics and Dynamical Astronomy*, 104(1), pp. 53–67. Available at: <https://doi.org/10.1007/s10569-009-9195-4>.
- Moulton, F.R. (1920) *Periodic Orbits*. Washington, Carnegie Institution of Washington.
- Roberts, G.E. (2002) “Linear Stability of the Elliptic Lagrangian Triangle Solutions in the Three-Body Problem,” *Journal of Differential Equations*, 182(1), pp. 191–218. Available at: <https://doi.org/10.1006/jdeq.2001.4089>.
- Zagouras, C.G. (1985) “Three-Dimensional Periodic Orbits about the Triangular Equilibrium Points of the Restricted Problem of Three Bodies,” *Celestial mechanics*, 37(1), pp. 27–46. Available at: <https://doi.org/10.1007/BF01230339>.